

Server Access logs and Storage Classes.

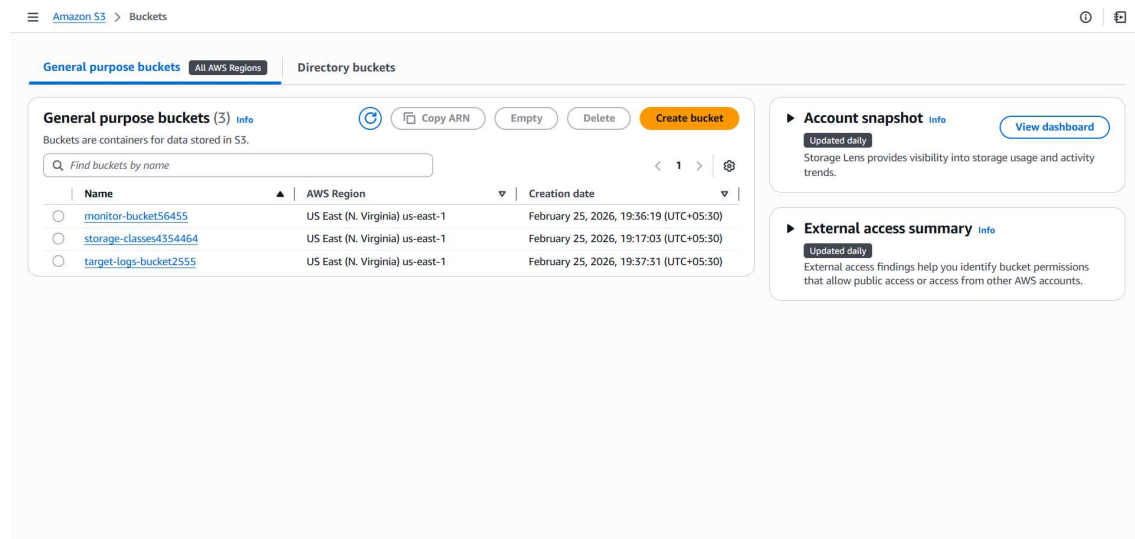
Server Access Logs in S3

In Amazon S3, **Server Access Logging** records detailed requests made to your bucket.

That means:

- Who accessed your bucket
- What object was accessed
- When it happened
- From which IP
- Whether it succeeded or failed

I have created two new buckets for the Server Access Logs.



One is target and the other one is source. In the properties tab of the source bucket we have to link the log file to the target bucket (works when both the buckets in same region).

Amazon S3 > Buckets > monitor-bucket56455 > Edit server access logging

☒ Enable

Bucket policy will be updated

When you enable server access logging, the S3 console automatically updates your bucket policy to include access to the S3 log delivery group.

Destination
Specify a destination bucket in the US East (N. Virginia) us-east-1 Region. To store your logs under a particular prefix, make sure that you include a slash (/) after the name of the prefix. Otherwise, the prefix will be added to the name of your log files.

s3://target-logs-bucket2555 [Browse S3](#)

Format: s3://<bucket>/<optional-prefix-with-path>

Destination Region
US East (N. Virginia) us-east-1

Destination bucket name
target-logs-bucket2555

Destination prefix
-

Log object key format

☐ [DestinationPrefix][YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

☒ [DestinationPrefix][SourceAccountId]/[SourceRegion]/[SourceBucket]/[YYYY]/[MM]/[DD]/[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]
To speed up analytics and query applications, use this format.

Source of date used in log object key format

☒ S3 event time
The year, month, and day will be based on the timestamp of the S3 event in the file that's been delivered.

☐ Log file delivery time
The year, month, and day will be based on the time when the log file was delivered to S3.

Log object key example
051741041276/us-east-1/monitor-bucket56455/2026/07/01/2026-07-01-00-00-00-[UniqueString]

Choose destination ×

S3 Buckets

Buckets that are not in the same Region as your source bucket (Europe (Stockholm) eu-north-1) can't be chosen.

Buckets (1/2)

Find buckets by name

	Name	AWS Region	Creation date
☐	source-bucket33452	Europe (Stockholm) eu-north-1	February 24, 2026, 22:27:23 (UTC+05:30)
☒	target-bucket345524	Europe (Stockholm) eu-north-1	February 24, 2026, 22:27:43 (UTC+05:30)

[Cancel](#) [Choose destination](#)

We have to specify which format we want for logs

Amazon S3 > Buckets > source-bucket33452 > Edit server access logging

Destination
Specify a destination bucket in the Europe (Stockholm) eu-north-1 Region. To store your logs under a particular prefix, make sure that you include a slash (/) after the name of the prefix. Otherwise, the prefix will be added to the name of your log files.

s3://target-bucket345524/logs [Browse S3](#)

Format: s3://<bucket>/<optional-prefix-with-path>

Destination Region
Europe (Stockholm) eu-north-1

Destination bucket name
target-bucket345524

Destination prefix
logs

Log object key format

☐ [DestinationPrefix][YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

☒ [DestinationPrefix][SourceAccountId]/[SourceRegion]/[SourceBucket]/[YYYY]/[MM]/[DD]/[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]
To speed up analytics and query applications, use this format.

Source of date used in log object key format

☒ S3 event time
The year, month, and day will be based on the timestamp of the S3 event in the file that's been delivered.

☐ Log file delivery time
The year, month, and day will be based on the time when the log file was delivered to S3.

Log object key example
logs035949051780/eu-north-1/source-bucket33452/2026/07/01/2026-07-01-00-00-00-[UniqueString]

Then we have to wait for about 15 to 30 mins while bucket creating the log file.

Amazon S3 > Buckets > target-logs-bucket2555

target-logs-bucket2555 info

Objects (1)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
logs051741041276/	Folder	-	-	-

Amazon S3 > Buckets > target-logs-bucket2555 > logs051741041276/ > us-east-1/ > monitor-bucket56455/ > 2026/ > 02/ > 25/

25/

Objects (17)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
2026-02-25-00-00-00-1A46716F0D43570C	-	February 25, 2026, 21:01:04 (UTC+05:30)	5.2 KB	Standard
2026-02-25-00-00-00-4B06D86D088A68C5	-	February 25, 2026, 21:21:59 (UTC+05:30)	627.0 B	Standard
2026-02-25-00-00-00-541900B287C53007	-	February 25, 2026, 21:23:48 (UTC+05:30)	2.4 KB	Standard
2026-02-25-00-00-00-60FC920E30A6CC98	-	February 25, 2026, 21:24:42 (UTC+05:30)	615.0 B	Standard
2026-02-25-00-00-00-672ESAC8E828971F	-	February 25, 2026, 21:23:46 (UTC+05:30)	621.0 B	Standard
2026-02-25-00-00-00-7196624DA164AD00	-	February 25, 2026, 21:13:51 (UTC+05:30)	1.3 KB	Standard
2026-02-25-00-00-00-79394AD4417A2046	-	February 25, 2026, 21:06:54 (UTC+05:30)	621.0 B	Standard

Storage Classes in Amazon S3 (Theory)

In Amazon S3, a **storage class** defines the cost, availability, durability, and retrieval characteristics of stored objects.

All S3 storage classes provide **11 nines durability (99.999999999%)**, but they differ in:

- Storage cost
- Retrieval cost
- Data availability
- Access latency
- Minimum storage duration

The selection of storage class depends on how frequently the data is accessed and how quickly it must be retrieved.

1 S3 Standard

S3 Standard is designed for **frequently accessed data**.

Characteristics:

- High availability (99.99%)
- Low latency
- Multi-Availability Zone storage
- No retrieval fee

Use Cases:

- Web applications
- Mobile applications
- Real-time analytics
- Frequently accessed APIs

It is the default storage class in S3.

2 S3 Intelligent-Tiering

S3 Intelligent-Tiering automatically moves objects between different access tiers based on usage patterns.

Characteristics:

- Automatic cost optimization
- No retrieval charges for frequent and infrequent tiers
- Small monitoring fee per object
- Designed for unpredictable access patterns

Use Cases:

- Applications with unknown usage behavior
- Data lakes

- Large datasets with variable access frequency

This class removes the need for manual lifecycle management.

3 S3 Standard-Infrequent Access (Standard-IA)

S3 Standard-IA is designed for data that is accessed less frequently but requires rapid access when needed.

Characteristics:

- Lower storage cost than Standard
- Retrieval fee applies
- Multi-AZ storage
- Minimum storage duration: 30 days

Use Cases:

- Backup files
- Disaster recovery data
- Monthly or quarterly reports

It balances cost savings with high availability.

4 S3 One Zone-Infrequent Access (One Zone-IA)

S3 One Zone-IA stores data in a single Availability Zone.

Characteristics:

- Lower cost than Standard-IA
- Retrieval fee applies
- Data stored in one AZ only
- Minimum storage duration: 30 days

Use Cases:

- Secondary backups
- Re-creatable data
- Test or development data

It offers reduced cost but lower resilience compared to multi-AZ classes.

5 S3 Glacier Storage Classes

Glacier storage classes are designed for archival and long-term data retention.

a) Glacier Instant Retrieval

- Millisecond access
- Lower cost than Standard-IA
- Minimum storage duration: 90 days

b) Glacier Flexible Retrieval (formerly Glacier)

- Retrieval time: Minutes to hours
- Very low storage cost
- Minimum storage duration: 90 days

c) Glacier Deep Archive

- Retrieval time: 12–48 hours
- Lowest storage cost
- Minimum storage duration: 180 days

Use Cases:

- Compliance data
- Long-term archives
- Legal or regulatory storage

In the created bucket we have to change the Storage Classes in the Management option using the Lifecycle Configuration.

Amazon S3 > Buckets > storage-classes4354464 > Lifecycle configuration > Create lifecycle rule

Lifecycle rule configuration

Lifecycle rule name

Up to 255 characters

Choose a rule scope
☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

Apply to all objects in the bucket
 If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)
☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

- ☐ Transition current versions of objects between storage classes
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☐ Expire current versions of objects
- ☐ Permanently delete noncurrent versions of objects
- ☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

Review transition and expiration actions

We don't recommend transitioning objects less than 120 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 120 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions	Days after object creation	
Standard-IA	35	Remove
Glacier Instant Retrieval	60	Remove
Glacier Deep Archive	90	Remove

The integer value for Glacier Instant Retrieval must be at least 30 more than the value for Standard-IA.

The integer value for Glacier Deep Archive must be at least 90 more than the value for Glacier Instant Retrieval.

[Add transition](#)

These are the review of the Storage class you have been creating.

Review transition and expiration actions

Current version actions

Day 0

- Objects uploaded

↓

Day 35

- Objects move to Standard-IA

↓

Day 60

- Objects move to Glacier Instant Retrieval

↓

Day 90

- Objects move to Glacier Deep Archive

Noncurrent versions actions

Day 0

No actions defined.

[Cancel](#) [Create rule](#)

Above are the steps to create the Storage Classes.

Amazon S3 > Buckets > storage-classes4354464 > Lifecycle configuration

The rule "mylifecyclerule" has been successfully added and the lifecycle configuration has been updated

It may take some time for the configuration to be updated. Refresh the lifecycle rules list if changes to the configuration aren't displayed.

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions

All storage classes 128K

Lifecycle rules (1)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

< 1 >

Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent versions...	Expired object delet...	Incomplete multipa...
mylifecyclerule	Enabled	Entire bucket	Transition to Standard-IA, th	-	-	-

The rule have been created successfully.